

# **WINNER ENGINEERING LEARNING HUB SERIES**

FUNDAMENTAL TOPIC

PIPING ACCESSORIES

& PROVISION

CONSIDERATION



Winner Engineering Pte Ltd  
永胜工程私人有限公司  
Excellence in Service

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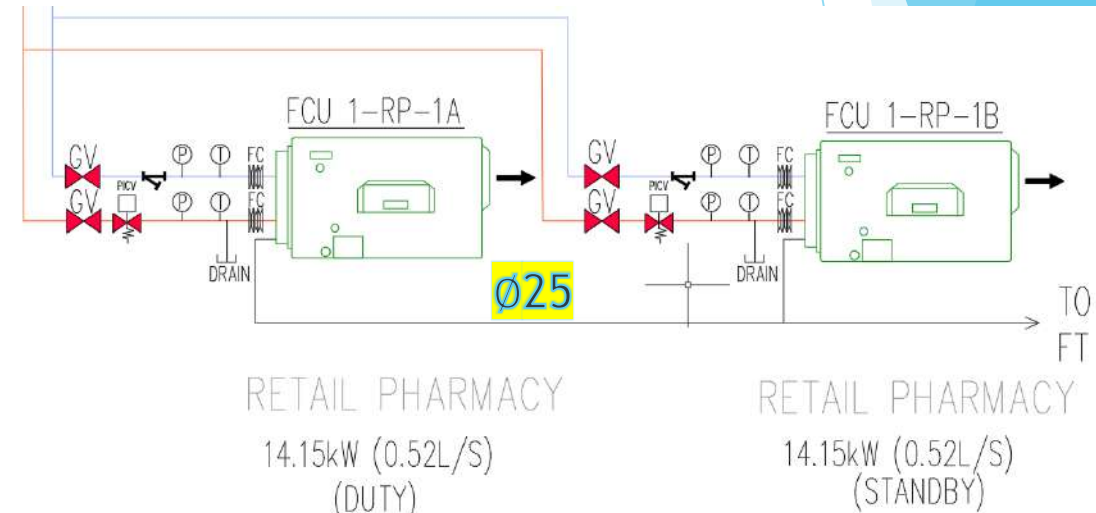
# condensate drain pipe

All drain pipes, gradient make sure to have 1:100 or 1% for smooth flow of water.

## condensate drain pipe sizing

TABLE OF CONDENSATE DRAIN PIPE

| COOLING CAP OF FCU | PIPE SIZES |
|--------------------|------------|
| BELOW 15 kW        | DIA 25     |
| 15.1kW TO 35kW     | DIA 32     |
| 35.1kW TO 175kW    | DIA 38     |
| ABOVE 175kW        | DIA 50     |



**Condensate drain pipe can be installed in two ways**

1. Exposed
2. concealed

# 1.EXPOSED DRAIN PIPE:



- Drain pipe hanged above the ceiling or run along the wall exposed.

**DRAIN PIPE**

# 2.CONCEALED DRAIN PIPE:



**FLOOR TRAP (OTHERS)**

- concealed drain pipes installed inside the concrete wall ,floor and brick wall.

**DRAIN PIPE**

**CONCRETE FLOOR**

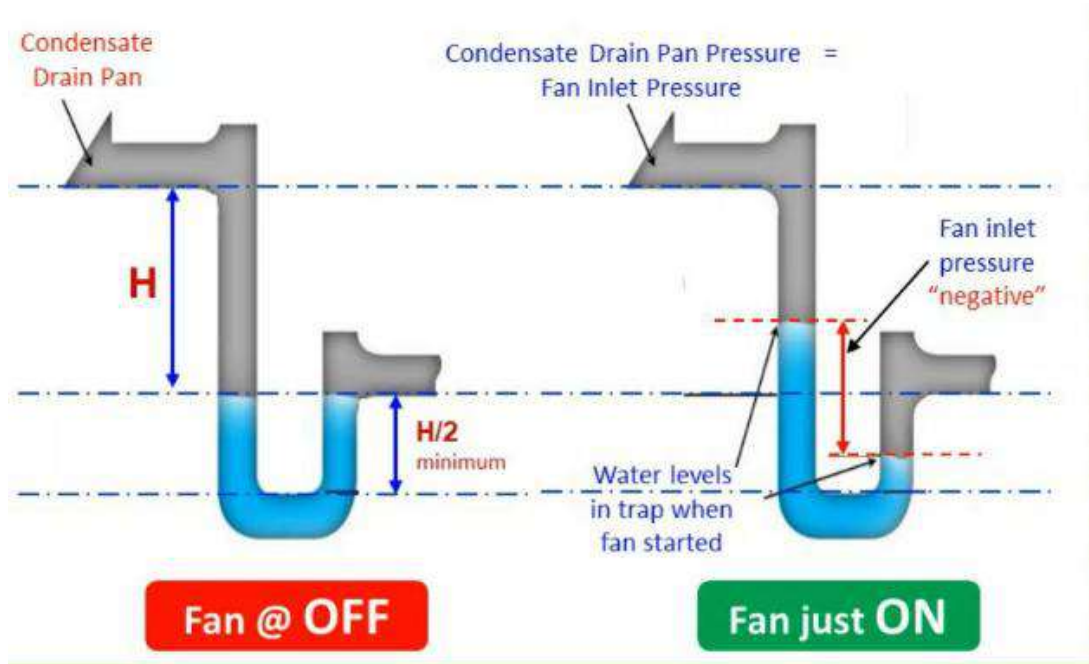
# Drain Pipe Insulation

As per the insulation of the drain pipe , 3/8”(10mm) thick closed- cell insulations are usually sufficient to prevent condensation. if our drain pipe is exposed to high temperatures (more the 30 °C), we should use 1/2” (13mm) thickness instead.



# Drain pipe Accessories & Provision:

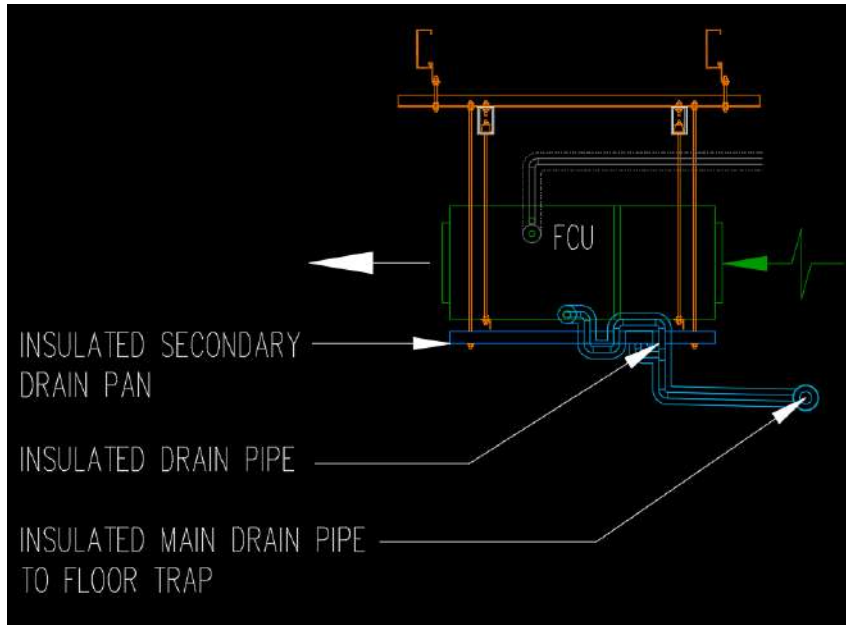
## U trap:



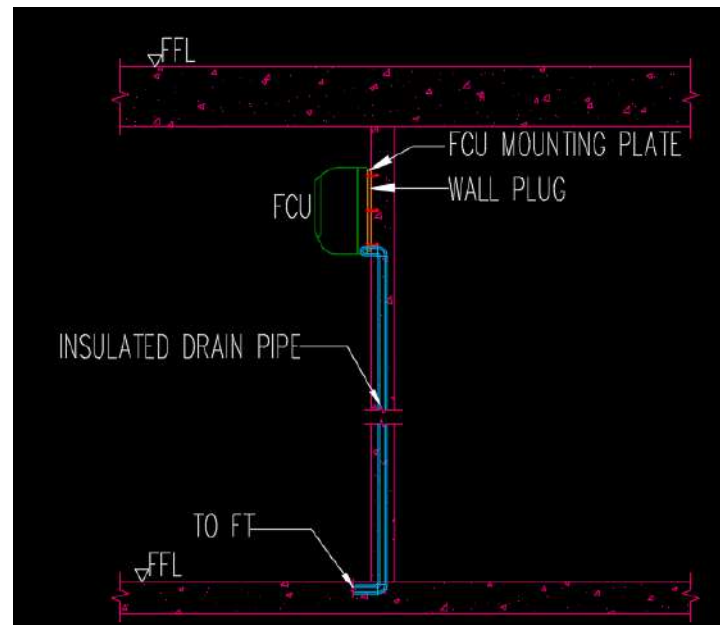
## Purpose of installing U- Trap:

1. Main purpose is stop sewer gases/bad odour/ bad smell back to the unit and space
2. Prevent the water back flow
3. Prevent the penetrations of any insects.

### DUCTED FCU DRAIN CONNECTION:



### WALL MOUNTED FCU DRAIN CONNECTION:

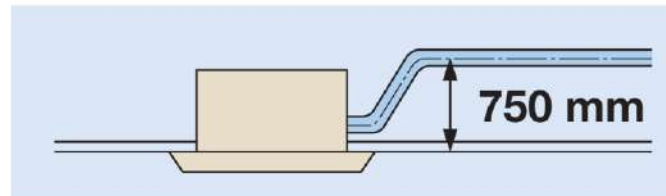


### FLOOR MOUNT AHU DRAIN CONNECTION:



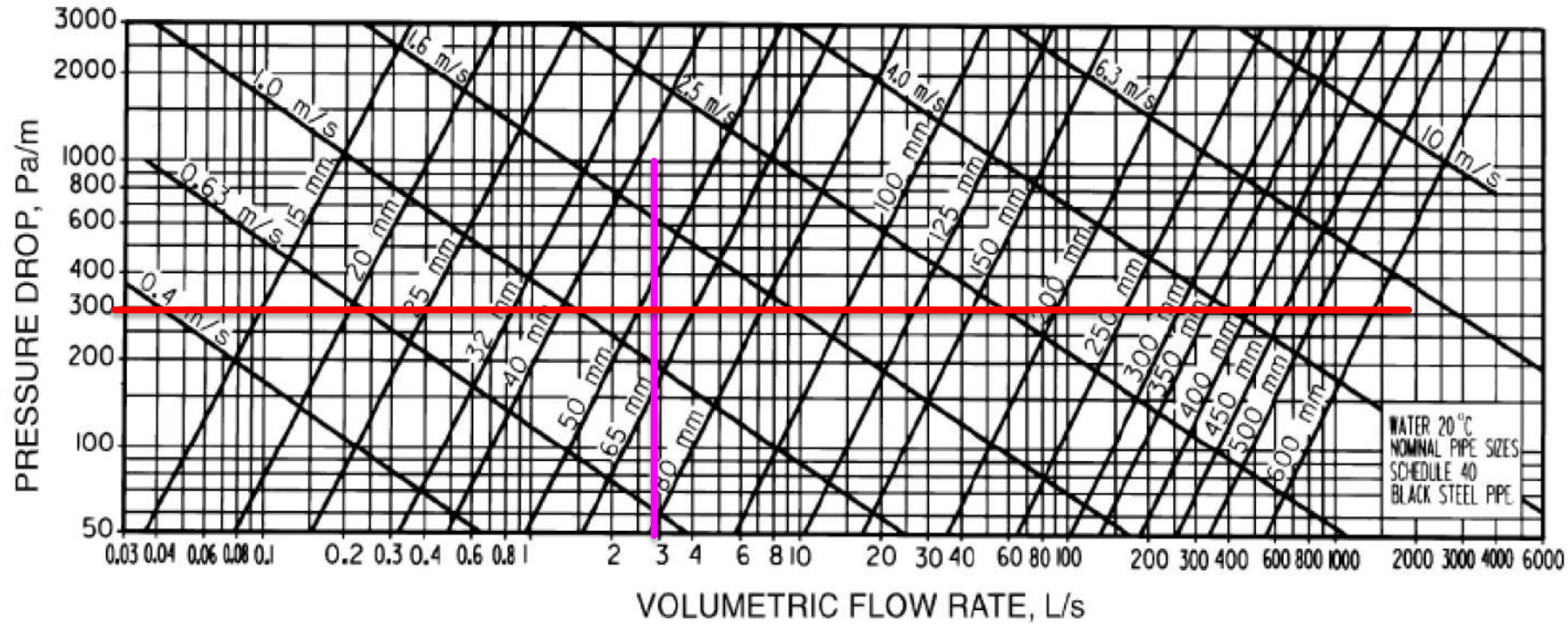
### CEILING CASSETTE FCU DRAIN CONNECTION:

- Drain pump is equipped as standard accessory with 750 mm lift.



# CHILLED WATER PIPE SIZEING:

**Table 1:** Friction Loss for Water in Commercial Steel Pipe (Scheduled 40)



**SCHEDULE OF CHILLED WATER AIR HANDLING UNITS**

| Specified Capacity |          | Offered Capacity |          | On-Coil |       | Off-Coil |       | Chilled Water |      | Water Flow Rate | Water Pipe Size | Water Pressure Drop | Supply Air |         |           | Coil Face Velocity |
|--------------------|----------|------------------|----------|---------|-------|----------|-------|---------------|------|-----------------|-----------------|---------------------|------------|---------|-----------|--------------------|
| Total              | Sensible | Total            | Sensible | DB      | WB    | DB       | WB    | In            | Out  |                 |                 |                     | Specified  | Offered |           |                    |
| (kw)               | (kw)     | (kw)             | (kw)     | (°C)    | (°C)  | (°C)     | (°C)  | (°C)          | (°C) | (l/s)           | (Ø mm)          | (kPa)               | (cmh)      | (cmh)   | (l/s)     | (m/s)              |
| 73                 | 49       | 79.78            | 56.33    | 23.10   | 17.20 | 10.30    | 10.28 | 8.0           | 15.0 | 2.72            | 65              | 46.73               | 12970      | 12970   | 3602.7778 | 1.4                |
| 88                 | 63       | 95.42            | 72       | 23.30   | 16.90 | 10.29    | 10.27 | 8.0           | 15.0 | 3.25            | 65              | 32.05               | 16300      | 16300   | 4527.7778 | 1.37               |
| 77                 | 41       | 84.24            | 45.59    | 22.10   | 18.40 | 10.39    | 10.39 | 8.0           | 15.0 | 2.87            | 65              | 49.34               | 11470      | 11470   | 3186.1111 | 1.3                |
| 97                 | 52       | 107.33           | 57.69    | 22.20   | 18.50 | 10.34    | 10.34 | 8.0           | 15.0 | <div>✓</div>    | 65              | 37.95               | 14340      | 14340   | 3983.3333 | 1.26               |
| 73                 | 39       | 80.51            | 43.18    | 22.00   | 18.40 | 10.36    | 10.36 | 8.0           | 15.0 | <div>✓</div>    | 65              | 45.52               | 10930      | 10930   | 3036.1111 | 1.24               |
| 105                | 57       | 115.8            | 63.21    | 22.00   | 18.30 | 10.49    | 10.49 | 8.0           | 15.0 | 3.94            | 65              | 29.65               | 16190      | 16190   | 4497.2222 | 1.28               |

# CHILLED WATER PIPE INSULATION SIZEING:

## Available Insulation Thickness

GLENSUL® insulation is available in various thicknesses. The following shows the available range:

|           | Series 25   |                                  | Series 40   |                                  | Series 50   |                                  | Series 65   |                                  | Series 75   |                                  |
|-----------|-------------|----------------------------------|-------------|----------------------------------|-------------|----------------------------------|-------------|----------------------------------|-------------|----------------------------------|
| Pipe Size | Casing Size | Approximate Insulation thickness | Casing Size | Approximate Insulation thickness | Casing Size | Approximate Insulation thickness | Casing Size | Approximate Insulation thickness | Casing Size | Approximate Insulation thickness |
| (mm)      | (mm)        | (mm)                             | (mm)        | (mm)                             | (mm)        | (mm)                             | (mm)        | (mm)                             | (mm)        | (mm)                             |
| 15        | 80          | 29                               | 100         | 39                               | 125         | 52                               | 150         | 64                               | 170         | 74                               |
| 20        | 80          | 27                               | 100         | 37                               | 125         | 49                               | 150         | 62                               | 170         | 72                               |
| 25        | 90          | 28                               | 115         | 41                               | 140         | 53                               | 160         | 63                               | 190         | 78                               |
| 32        | 90          | 24                               | 115         | 36                               | 140         | 49                               | 170         | 64                               | 190         | 74                               |
| 40        | 100         | 26                               | 125         | 38                               | 140         | 46                               | 170         | 61                               | 200         | 76                               |
| 50        | 115         | 27                               | 140         | 40                               | 160         | 50                               | 190         | 65                               | 215         | 77                               |
| 65        | 125         | 25                               | 150         | 37                               | 170         | 47                               | 200         | 62                               | 224         | 74                               |
| 80        | 140         | 26                               | 170         | 41                               | 190         | 51                               | 215         | 63                               | 240         | 76                               |
| 100       | 170         | 28                               | 190         | 38                               | 215         | 50                               | 240         | 63                               | 265         | 75                               |
| 125       | 190         | 25                               | 224         | 42                               | 240         | 50                               | 265         | 63                               | 295         | 78                               |
| 150       | 215         | 25                               | 240         | 37                               | 265         | 50                               | 295         | 65                               | 320         | 77                               |
| 200       |             |                                  | 295         | 38                               | 320         | 50                               | 350         | 65                               | 375         | 78                               |
| 250       |             |                                  | 350         | 39                               | 375         | 51                               | 400         | 64                               | 425         | 76                               |
| 300       |             |                                  | 400         | 38                               | 425         | 51                               | 450         | 63                               | 475         | 76                               |
| 350       |             |                                  | 425         | 35                               | 450         | 47                               | 475         | 60                               | 510         | 77                               |
| 400       |             |                                  | 475         | 34                               | 510         | 52                               | 530         | 62                               | 560         | 77                               |
| 450       |             |                                  | 530         | 37                               | 560         | 52                               | 590         | 67                               | 610         | 77                               |
| 500       |             |                                  | 590         | 41                               | 610         | 51                               | 635         | 64                               | 660         | 76                               |
| 600       |             |                                  |             |                                  | 710         | 50                               | 735         | 63                               | 760         | 75                               |
| 700       |             |                                  |             |                                  | 800         | 45                               | 850         | 70                               | 900         | 95                               |
| 800       |             |                                  |             |                                  | 900         | 44                               | 950         | 69                               | 1000        | 94                               |
| 900       |             |                                  |             |                                  | 1000        | 43                               | 1050        | 68                               | 1100        | 93                               |
| 1000      |             |                                  |             |                                  | 1100        | 42                               | 1150        | 67                               | 1200        | 92                               |

■ Recommended insulation thickness under normal operating condition.

## ONSITE INSTALLTION



# Chilled water pipe Accessories & Provision:



- ❑ Flexible connector
- ❑ Strainer
- ❑ Motorized valve
- ❑ Gate valve
- ❑ Balancing valve
- ❑ Temperature Gauge
- ❑ Drain cock
- ❑ Air vent

# Chilled water pipe Accessories & Provision:

2 way  
Motorized Valve



Strainer



Pressure Gauge



Flexible Connector



Balancing Valve



Gate Valve



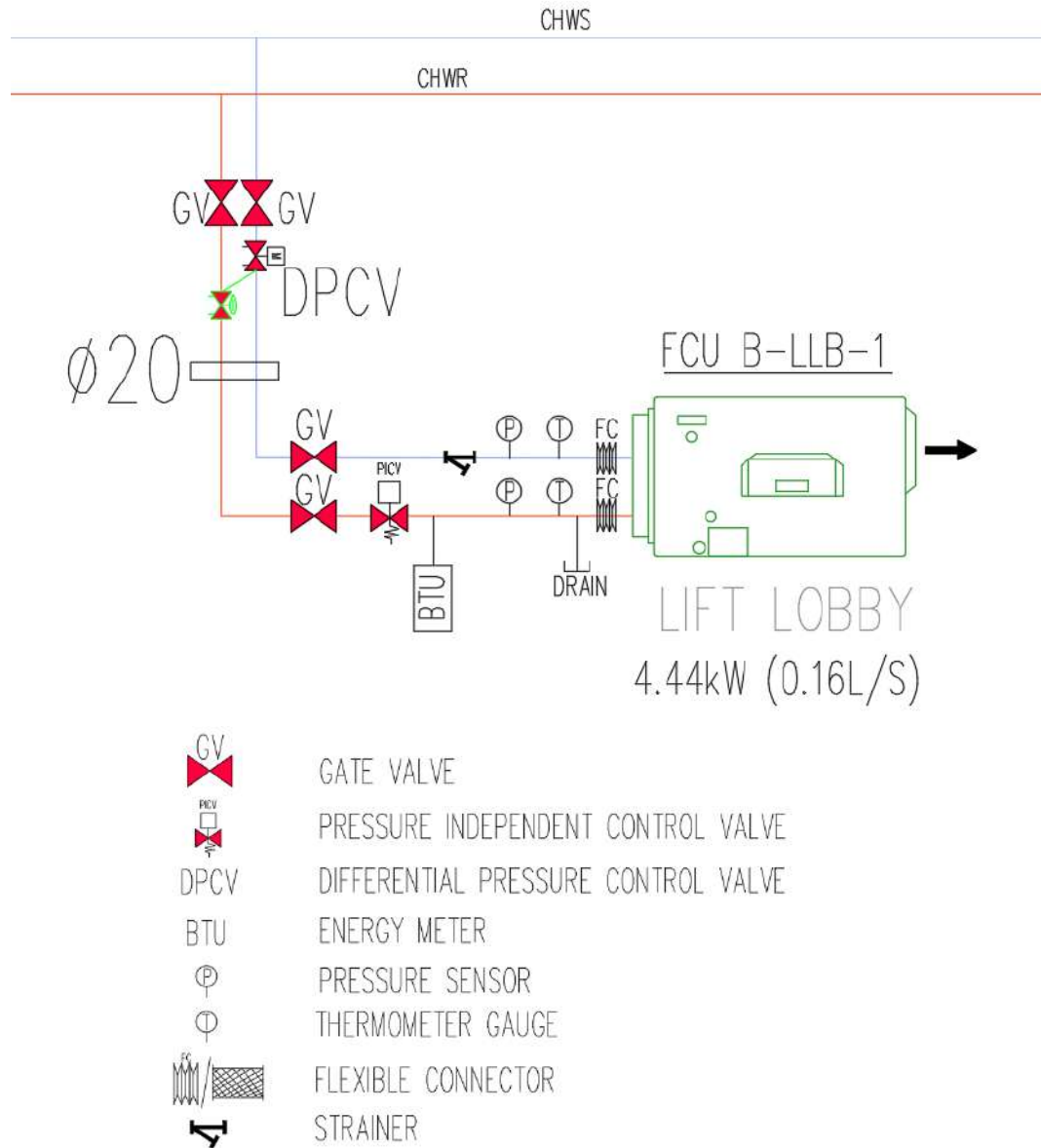
Thermometer



Chilled water Pipe with Jacketing



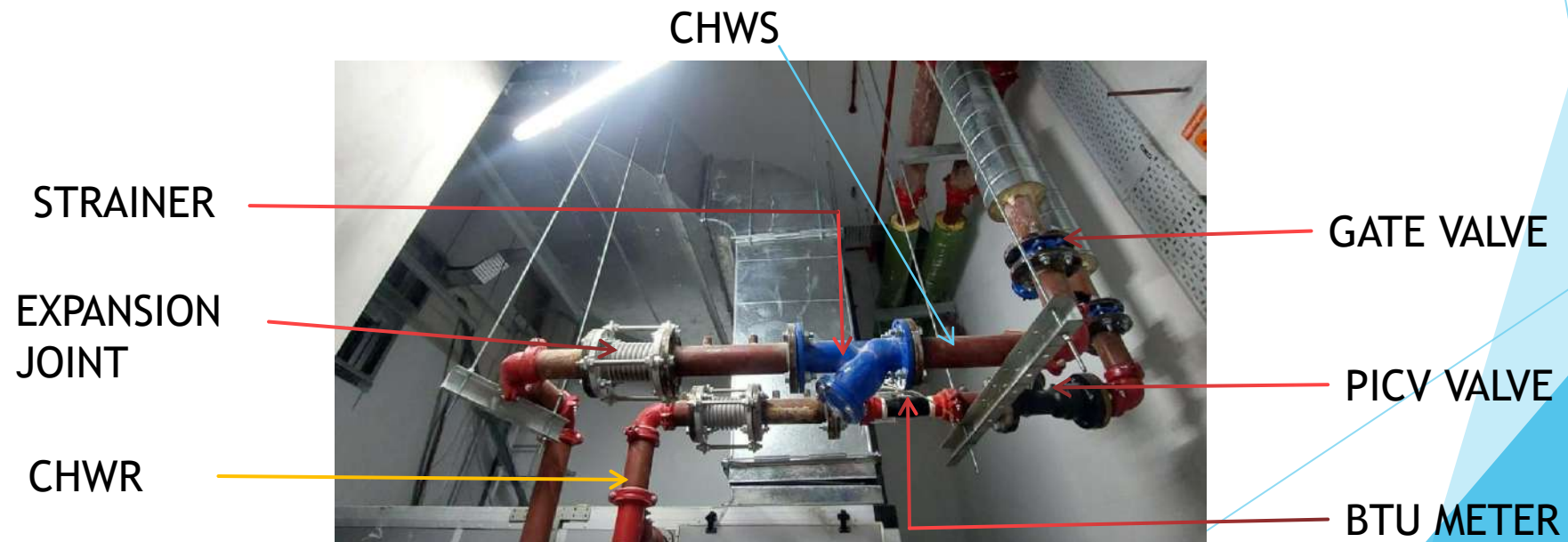
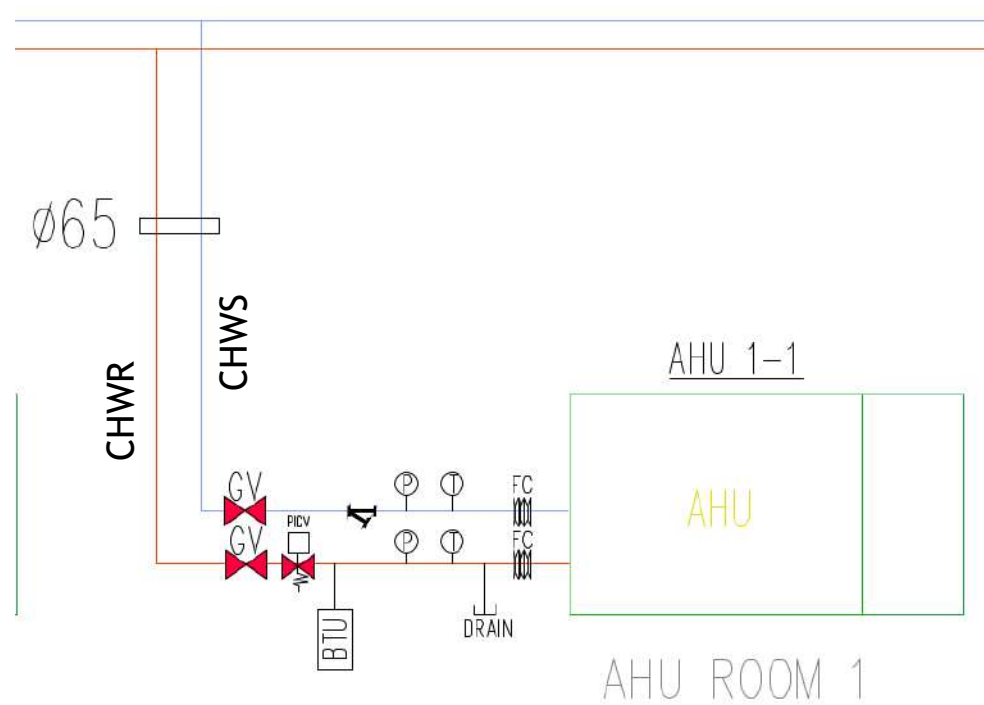
# Chilled water pipe Accessories & Provision:



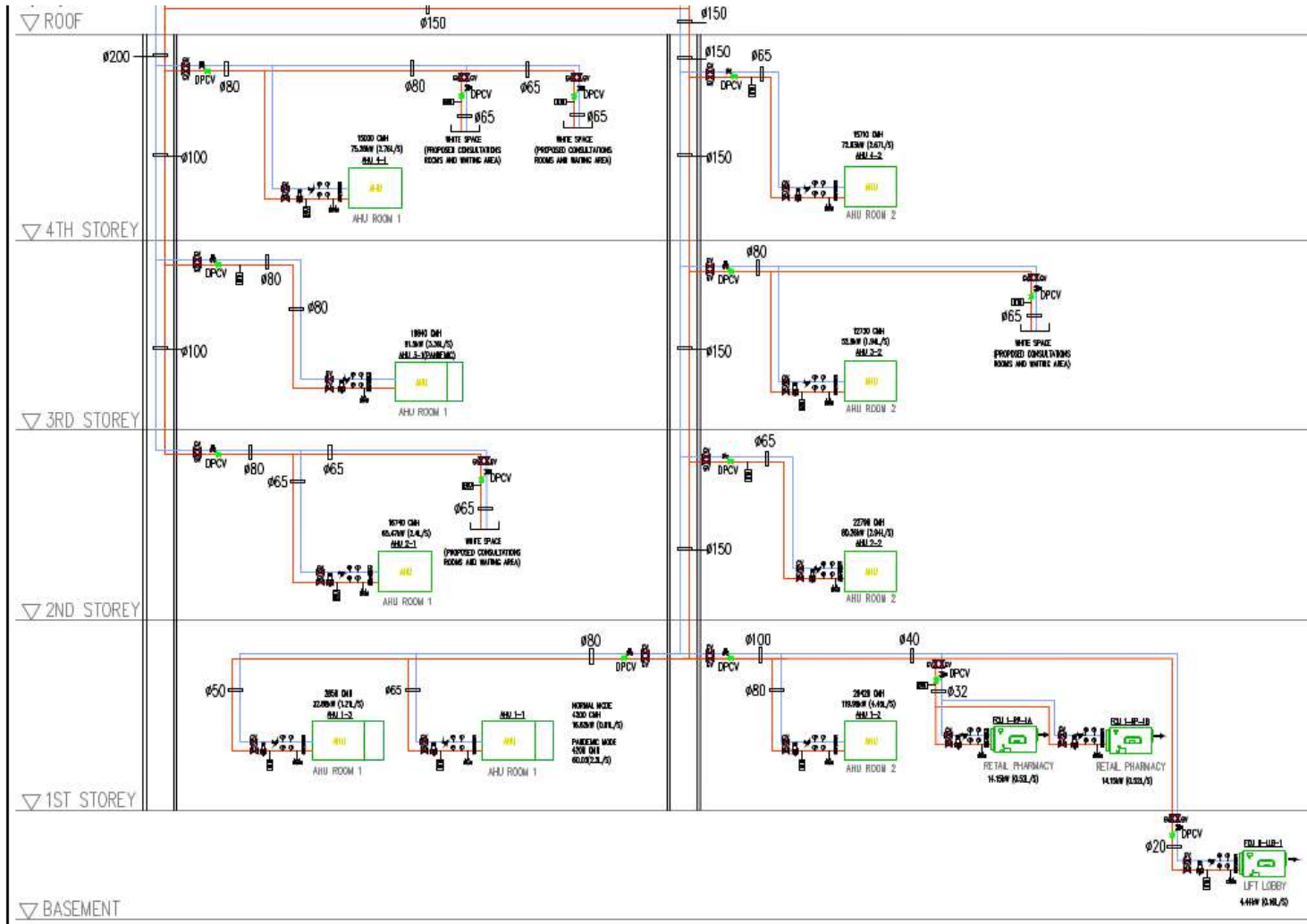
## PICV VALVE:

Pressure independent control valves are an automatic temperature control valve and an automatic flow regulating valve packaged in one valve body.

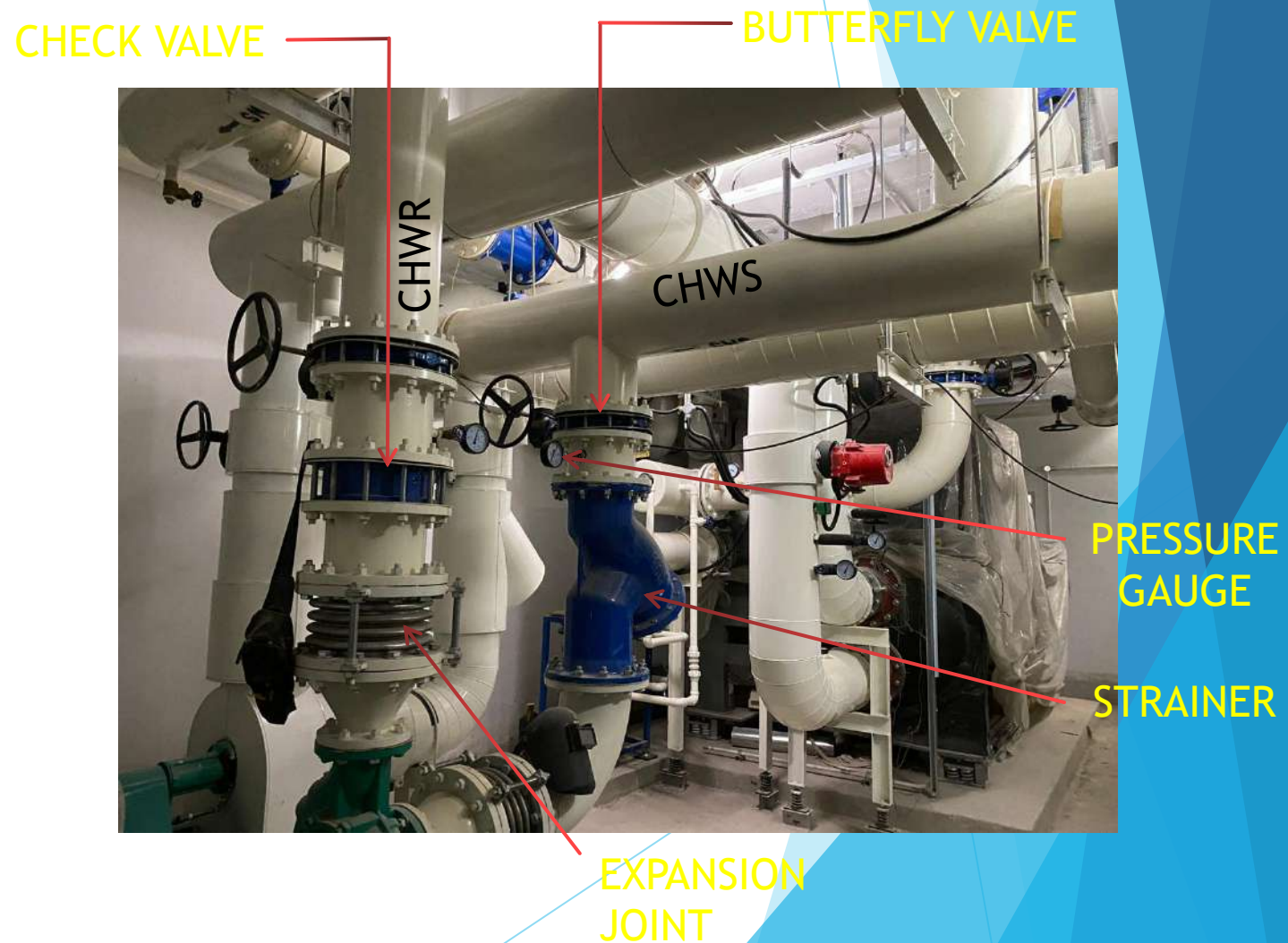
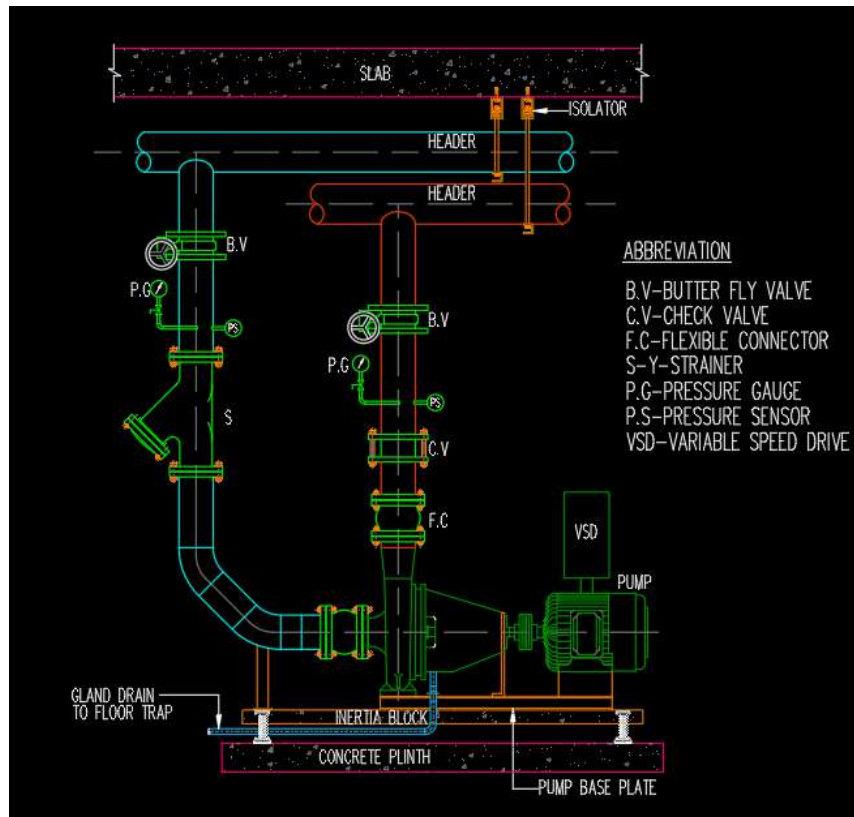
# TYPICAL AHU CONNECTION:



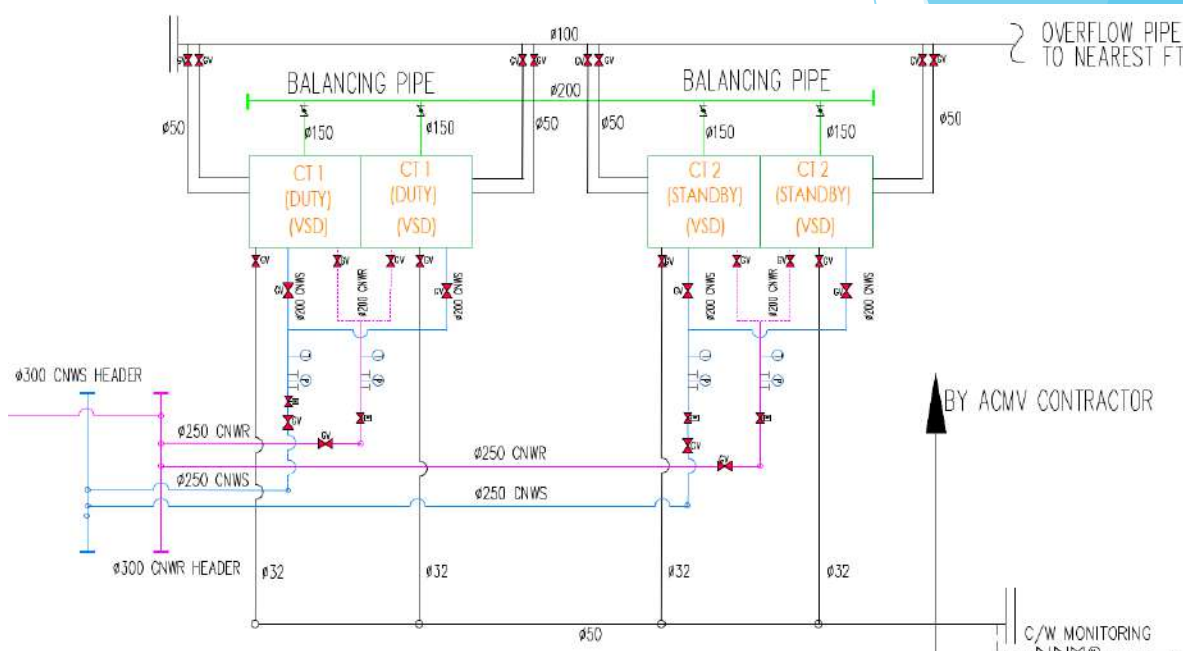
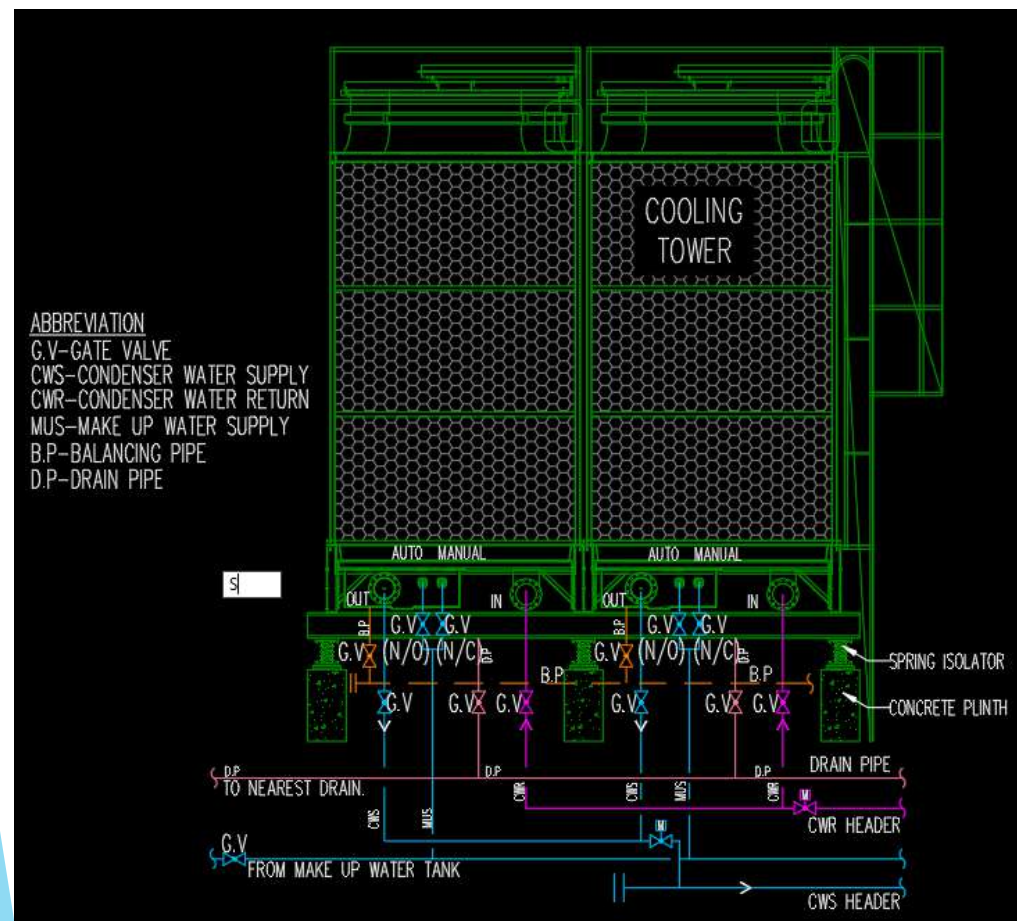
# SCHEMATIC:



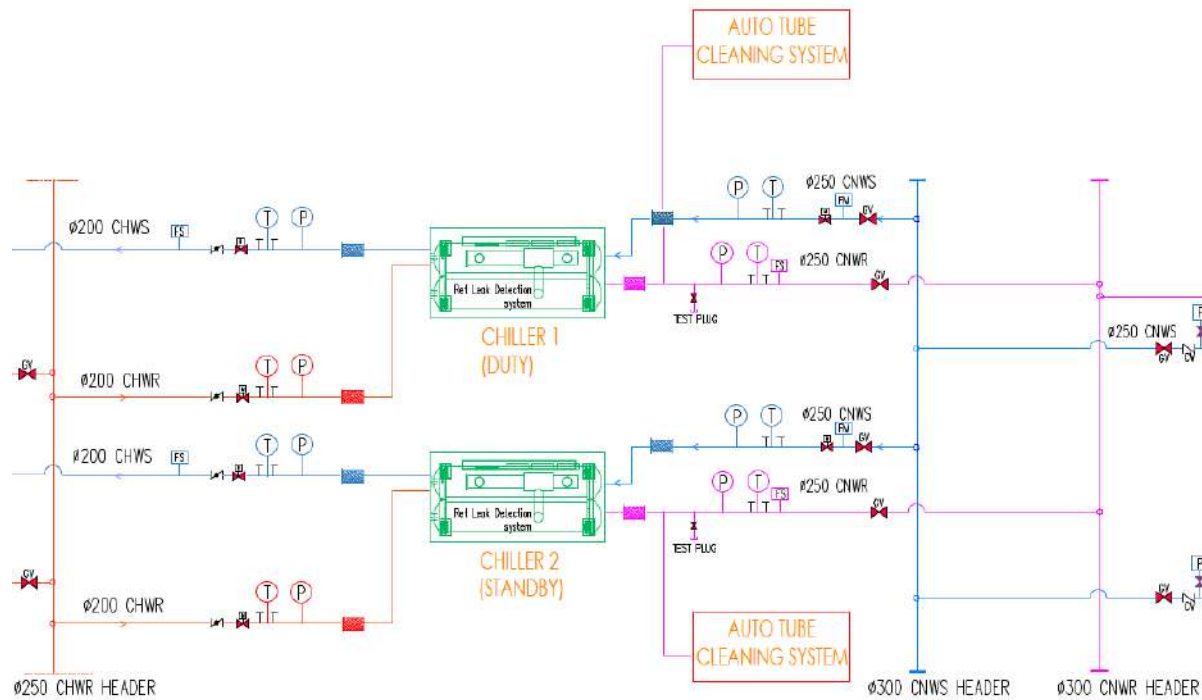
## TYPICAL PUMP CONNECTION:



# TYPICAL COOLING TOWER CONNECTION:



# TYPICAL CHILLER CONNECTION:



BUTTERFLY VALVE

MOTORIZED VALVE

